

Next.js handles JS and CSS minification **automatically** during the production build process ( `next build` ). You generally don't need to configure anything for it to work out of the box.  Next.js +3

## How it Works

- **JavaScript Minification:** Next.js uses the [SWC compiler](#), a high-performance tool written in Rust. It is significantly faster than traditional tools like Terser.
- **CSS Minification:** In production, Next.js automatically [concatenates and minifies](#) all CSS files (including CSS Modules) to ensure the smallest possible payload is delivered to the browser.  Next.js +3

## Key Configuration Options

While minification is enabled by default, you can manage it via `next.config.js` : 

- **SWC Minification:** Since Next.js 13, `swcMinify` is enabled by default. If you are on an older version or need to explicitly set it, you can use:

### javascript

```
// next.config.js
module.exports = {
  swcMinify: true, // Uses SWC instead of Terser for 7x faster minification
}
```

Use code with caution.

- **Disabling Minification (Not Recommended):** If you need to disable minification for debugging production builds, you can override the [Webpack configuration](#):

### javascript

```
// next.config.js
module.exports = {
  webpack: (config, { dev }) => {
    if (!dev) {
      config.optimization.minimize = false;
    }
    return config;
  }
}
```

```
},  
};
```

Use code with caution.



## Common Troubleshooting

- **Minification Errors:** If your build fails with "CSS minification error" or "Unclosed block," it is often due to syntax errors in your CSS or issues within a third-party library.
- **Development vs. Production:** Minification **only occurs during the production build** ( `next build` ). In development mode ( `next dev` ), files are left unminified to support Fast Refresh and easier debugging.
- **Gzip Compression:** In addition to minification, [Next.js](#) provides [Gzip compression](#) by default for static files to further reduce transfer size.  Next.js +4